Crypto Privacy Roadmap

Durable systems understanding: how privacy in cryptocurrency works, where it fails, and how to use it safely.

Core filter. Learn mechanisms, not products. Wallets, coins and mixers change every few years. The ideas they are built from do not: public ledgers leak, keys are the whole security model, metadata outlives content, and trust never disappears, it only moves. The target is that you can look at any crypto system, including one that does not exist yet, and predict what it reveals and where it can fail before you use it.

Math is minimal. Treat each cryptographic primitive as a black box with a stated guarantee. Learn the conditions under which the guarantee breaks.

How each phase is laid out. Learn (the concepts), Study (specific chapters and one primary link per source, no link piles), Milestone (a hands-on test of the concept).

Compiled 8 November 2026. Links and chapter numbers were checked against published tables of contents and web search results on this date. See the Source notes page for what was and was not verified.

PHASE 1

Privacy, Trust and Threat Models

LEARN

- Four separate properties: confidentiality (content hidden), anonymity (identity hidden), unlinkability (actions cannot be connected), metadata privacy (who, when, how much).
- Adversary models: a data broker, an exchange, a thief, a competitor, a state. A defence that beats one often does nothing against another.
- Trust assumptions: for every tool, ask who you must trust and what they could do with that trust.
- Privacy (hiding information from parties with no right to it) is a different goal from evasion of legal obligations. Keep them separate from the start.

STUDY

- EFF Surveillance Self-Defense, module Your Security Plan. Read it fully; it is the standard threat-modelling worksheet. Then Seven Steps To Digital Security.
- Bitcoin Wiki, Privacy: read only the introduction and overview to see how threats are categorised (on-chain vs off-chain). The page is long; use it as a reference, not cover to cover.
- Optional: Michael Bazzell, *Extreme Privacy* (book), for compartmentalisation. No chapter mapping given because I could not verify the current edition's contents.

Milestone. Write a one-page threat model for a hypothetical holder using EFF's questions: what do I want to protect, who from, how bad is failure, how likely is the threat, how much effort will I accept. Revise it after each later phase.

PHASE 2

Cryptographic Primitives as Black Boxes

LEARN

- Hash functions: fingerprints, tamper detection.
- Key pairs and digital signatures: what a signature proves and what it does not.
- Randomness: why bad entropy destroys everything built on it.
- Commitments and zero-knowledge proofs: concept only (prove a claim without revealing the data).
- Common failures: reused nonces, weak randomness, leaked or poorly stored keys, home-made crypto.

STUDY

- Aumasson, *Serious Cryptography* (publisher page; Chapter 2 is a free download): **Ch. 2 Randomness**, **Ch. 6 Hash Functions**, **Ch. 12 Elliptic Curves** (read the signing section and the "How Things Can Go Wrong" parts, including ECDSA with bad randomness). Skip the heavy math. The 2nd edition (2024) adds **Ch. 15 Cryptocurrency Cryptography**.
- Antonopoulos & Harding, *Mastering Bitcoin*, 3rd ed. (free on GitHub): **Ch. 4 Keys and Addresses**, **Ch. 8 Signatures**.

Milestone. Using a vetted library, generate a key pair, sign a message, verify it, change one bit, and watch verification fail. Hash a file, change one byte, compare. Never implement the primitives yourself for real use.

PHASE 3

How a Ledger Actually Works

LEARN

- Transaction models: UTXO (Bitcoin) vs account-based (Ethereum). They leak different things.
- Nodes, mempool, blocks, consensus: where your transaction goes and who sees it.
- A wallet is a key manager, not a container of coins. Seed phrases, deterministic derivation, address types.
- Full node vs light client: what each verifies and what each reveals.

STUDY

- *Mastering Bitcoin*, 3rd ed.: **Ch. 1 Introduction**, **Ch. 2 Overview**, **Ch. 3 Bitcoin Core**, **Ch. 5 Wallets**, **Ch. 6 Transactions**, **Ch. 11 Bitcoin Blockchain** (covers signet, a test network). Appendix A contains the Bitcoin whitepaper.
- Ethereum documentation: Ethereum accounts for the account model, externally owned vs contract accounts, and why an account is not a wallet.

Milestone. On a test network, build and broadcast a transaction. Decode the raw transaction and label every field. List what a stranger can learn from it permanently.

PHASE 4

Why Public Ledgers Leak

LEARN

- Pseudonymous is not anonymous: every transaction is public forever.

- Address clustering: inputs spent together are assumed to share an owner, change-output detection, address reuse.
- Entry and exit points: identity-verified exchanges link addresses to people.
- Permanence: analysis improves over time, so old mistakes get more revealing, not less.
- How chain-analysis firms work, and what they cannot do.

STUDY

- Meiklejohn et al., *A Fistful of Bitcoins*: start with the short USENIX ;login: version, then the full IMC 2013 paper. This is the foundational clustering study.
- Bitcoin Wiki Privacy, the section on blockchain explorer websites and the "bad privacy example" sections. The article dates from 2019 and is about 21,000 words, so read selectively.

> **Milestone.** Pick a public transaction on a block explorer and trace it by hand across several hops. Note which heuristic you used at each step.

PHASE 5

Network Layer and Metadata

LEARN

- Broadcast leaks: the first node to see a transaction can often tie it to an IP address.
- Light wallets leak your addresses to a server.
- Tor and onion routing, and what a VPN does and does not do (it moves trust, it does not remove it).
- Timing and traffic correlation.

STUDY

- *Mastering Bitcoin*, 3rd ed., **Ch. 10 Bitcoin Network**: the 3rd edition expanded its discussion of the privacy problems of bloom filters and added compact block filters. Read both.
- Dingledine, Mathewson, Syverson, Tor: The Second-Generation Onion Router (2004). Read the design goals and threat model. Note what the authors say Tor does not protect against, including end-to-end timing and intersection attacks. The network has changed since 2004; the threat model has not.

> **Milestone.** Run a node and observe its traffic with a packet-capture tool, with and without Tor. Write down exactly what a network observer sees in each case.

PHASE 6

Privacy Mechanisms

LEARN

- Learn each as: problem, mechanism, what it hides, what it still leaks, known attacks.
- Coin mixing and CoinJoin: breaking input-output links, and why amounts and later behaviour matter.
- Monero: ring signatures (sender), stealth addresses (receiver), confidential amounts.
- Zcash: shielded transactions, and why the size of the actual shielded crowd limits the privacy.
- Layer-2 and Ethereum-side tools: stealth addresses, mixers, and their legal history.
- Cross-cutting concepts: anonymity set (the size of the crowd you hide in), fungibility, privacy by default vs opt-in.

STUDY

- *Zero to Monero*, 2nd ed. (free, Monero library): **Ch. 3** ring signatures (CLSAG in section 3.6), **Ch. 4** addresses, **Ch. 5** amount hiding, **Ch. 6** transactions. Published April 2020, so later protocol changes are not covered. Confirm current parameters in Monero's official documentation.
- Zcash: Electric Coin Company, How Transactions Between Shielded Addresses Work. A simplified concept introduction that says it omits details. Use it to understand the idea, then check current official documentation for today's shielded pools.
- CoinJoin: Bitcoin Wiki Privacy page, CoinJoin sections (no stable anchor verified; use the page's contents list).
- Ethereum-side tools: no stable primary reference verified, so none is listed. For any tool, read its own documentation and any published audit or legal ruling.
- Supplement: *Serious Cryptography* 2nd ed., Ch. 15 Cryptocurrency Cryptography.

Milestone. For each mechanism, write a half-page analysis titled "what an observer can still learn." Use test networks only. Then compare the analyses side by side.

PHASE 7

Keys, Devices and Operational Security

LEARN

- Key storage: hardware wallets, seed backups, single points of failure.
- Endpoint risks: malware, clipboard hijacking, phishing, fake wallet software, supply-chain compromise.
- Social and physical risk: publicised holdings attract targeting, and no cryptography stops coercion.
- Compartmentalisation: separate wallets and identities per purpose, no address reuse.
- Recovery and inheritance: a backup that no one can find or restore is a loss.

STUDY

- *Mastering Bitcoin*, 3rd ed., **Ch. 13 Security**.
- EFF Surveillance Self-Defense: the guides on avoiding phishing attacks, passwords and keeping your data safe (find them in the site index).
- Bitcoin Wiki Privacy: the non-blockchain threats sections.

Milestone. Do a full recovery drill on a test wallet. Wipe the device, restore from the seed on a clean device, confirm access. Then write a table of failure modes: what breaks, how you would notice, what you would lose.

PHASE 8

Rules and Law

LEARN

- This is the least durable phase. Verify it against current sources.
- Custody and taxation, identity verification at exchanges, reporting rules, sanctions law.
- Jurisdictions treat privacy tools very differently, and some have prosecuted tool developers or restricted privacy coins on exchanges.
- The law of your own country matters more than any tool choice.

STUDY

- FATF, Updated Guidance for a Risk-Based Approach to Virtual Assets and VASPs (October 2021): the international baseline behind exchange identity checks and the Travel Rule.
- FATF, Targeted Update on Implementation (June 2022). FATF has published later updates; check its site for the newest.
- Your own government's tax and financial-regulator publications. No link given because it depends on your country.

Milestone. Write a short summary of the rules that apply to you (tax reporting, exchange requirements, what is restricted), sourced only from your government's official publications.

WRAP-UP

What "Done" Looks Like

You can look at any crypto system and reason about:

- What it publishes permanently, and who can link it to a person.
- Where the trust sits (key holder, server, mixer coordinator, exchange).
- What a network observer, a chain analyst and a thief each see.
- How it fails, and what that failure costs.
- Why a privacy claim is true, partly true or marketing, before you use it.

Pacing

Phases 1 and 2 take about 2-3 weeks. Phases 3 and 4 take about 4-6 weeks and carry the most insight. The remaining phases take about 2 weeks each. At roughly 5 hours a week this is 4-5 months with overlap. Do the milestones on test networks or with trivial amounts. The failures you cause on purpose are the ones that teach.

REFERENCE

Reading List at a Glance

One primary source per topic. Titles and editions are given so you can find each item even if a link changes.

Source	Edition / date	Phases	Link
EFF Surveillance Self-Defense	Living site	1, 7	https://ssd.eff.org/
Mastering Bitcoin (Antonopoulos & Harding)	3rd ed., Dec 2023	2, 3, 5, 7	https://github.com/bitcoinbook/bitcoinbook/blob/develop/BOOK.md
Serious Cryptography (Aumasson)	1st ed. 2017; 2nd ed. 2024 adds Ch. 15	2, 6	https://nostarch.com/SeriousCrypto
Ethereum accounts documentation	Living page	3	https://ethereum.org/developers/docs/accounts/
A Fistful of Bitcoins (Meiklejohn et al.)	IMC 2013; ;login: 2013	4	https://www.usenix.org/publications/login/december-2013-volume-38-number-6/fistful-bitcoins-characterizing-payments-among
A Fistful of Bitcoins, full paper	IMC 2013	4	https://smeiklej.com/files/imc13.pdf
Bitcoin Wiki: Privacy (Chris Belcher)	2019, edited since	1, 4, 5, 6, 7	https://en.bitcoin.it/Privacy
Tor: The Second-Generation Onion Router	USENIX Security 2004	5	https://www.usenix.org/conference/13th-usenix-security-symposium/tor-second-generation-onion-router
Zero to Monero	2nd ed., April 2020 (v2.0.0)	6	https://web.getmonero.org/library/
How Transactions Between Shielded Addresses Work	Electric Coin Company blog; older	6	https://electriccoin.co/blog/zcash-private-transactions
FATF Updated Guidance, Virtual Assets and VASPs	October 2021	8	https://www.fatf-gafi.org/publications/fatfrecommendations/documents/guidance-rba-virtual-assets-2021.html
FATF Targeted Update on VAs and VASPs	June 2022	8	https://www.fatf-gafi.org/publications/fatfrecommendations/documents/targeted-update-virtual-assets-vasps.html

SOURCE NOTES

What Was Verified and What Was Not

Verified against published tables of contents (8 November 2026): *Mastering Bitcoin* 3rd ed. chapter list and the chapters the 3rd edition revised (keys, signatures, network, blockchain/signet); *Serious Cryptography* chapter numbers 2, 6 and 12, and Ch. 15 in the 2nd edition; *Zero to Monero* 2nd ed. structure (CLSAG in 3.6, addresses in Ch. 4, amount hiding in Ch. 5, transactions in Ch. 6).

Links: each URL appeared in web search results on the compile date. They were not opened in a browser session, so a page may have moved. If one fails, search the title and edition given in the table.

Not verified, so not mapped: chapter contents of Bazzell's *Extreme Privacy*; a canonical reference for Ethereum-side privacy tools; exact section anchors inside the Bitcoin Wiki Privacy page beyond the blockchain-explorer section; current Zcash and Monero protocol parameters.

Time-sensitive: the Monero book dates from 2020, the Zcash article is older and simplified, the Bitcoin Wiki article dates from 2019, and regulation (Phase 8) changes quickly. For anything that affects real money, confirm against current official documentation.

This roadmap is educational. It is not legal, tax or financial advice.